How to Spot a Misleading Graph



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Watch: How to Spot a Misleading Graph

Play on TED

TED-Ed lesson by **Lea Gaslowitz** · directed by Mark Phillips · narrated by Pen-Pen Chen (2017)

Lesson page with quiz & discussion: ed.ted.com/lessons/how-to-spot-a-misleading-graph-lea-gaslowitz

> *While you watch: note every trick the video shows. Which ones have you seen at work?*

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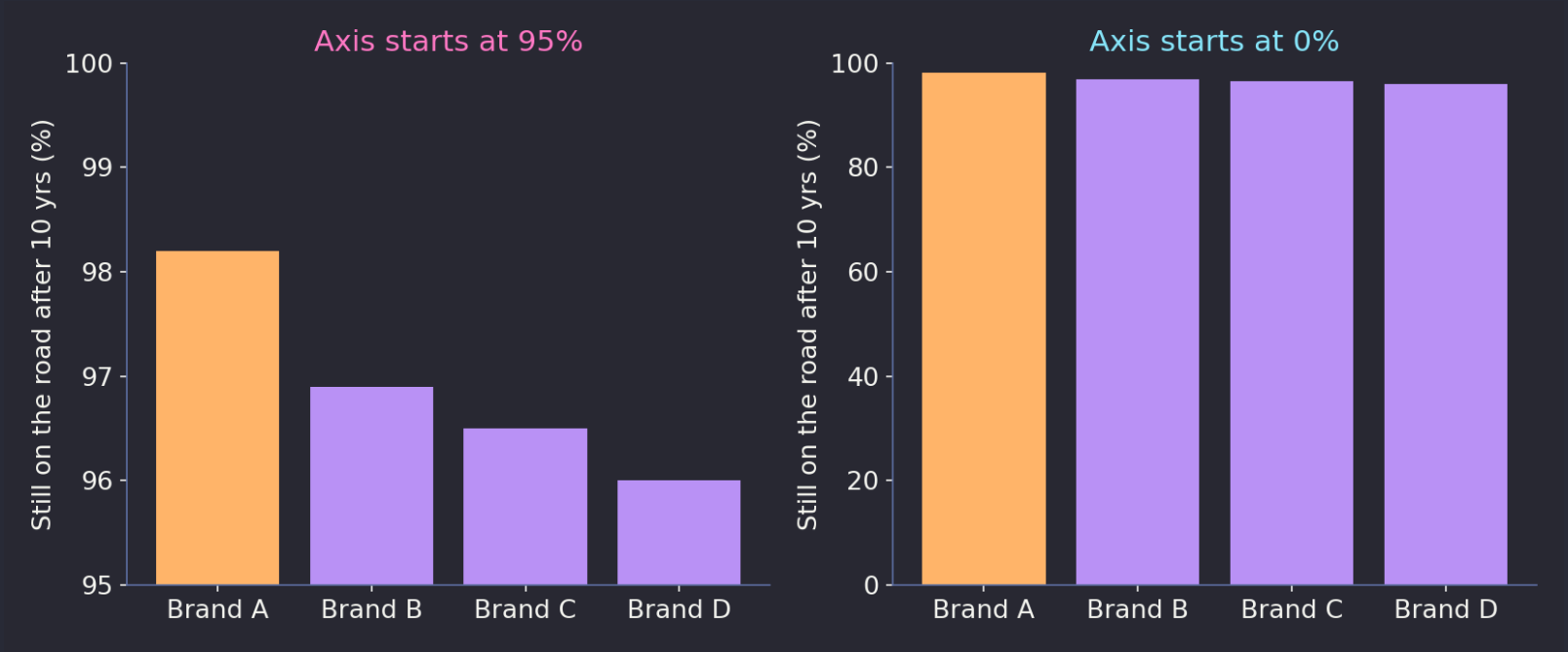
Why It Matters

- Graphs help us grasp complex data **at a glance**
- That same speed makes them easy to **misuse**
- Misleading graphs can be **deliberate** — or simply **careless**
- Either way, the reader walks away with the wrong conclusion

> *A chart is an argument. Check it before you accept it.*

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1. Check the Y-Axis — Where Does It Start?



Illustrative data. Same four numbers, two different y-axis ranges.

- A **truncated axis** turns a ~2-point gap into a "huge" lead
- Bar charts are worst: we read **bar length as the value**

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1. Check the X-Axis Too

- Are the intervals **evenly spaced**?
  - e.g. 2000, 2005, 2010, **2023** — a gap hidden in plain sight
- Is any period **skipped** or **squeezed**?
- Is the axis **labelled**, with **units**?
- Is any axis **missing a scale** altogether?

> *No scale = no way to judge size. Treat it as decoration, not evidence.*

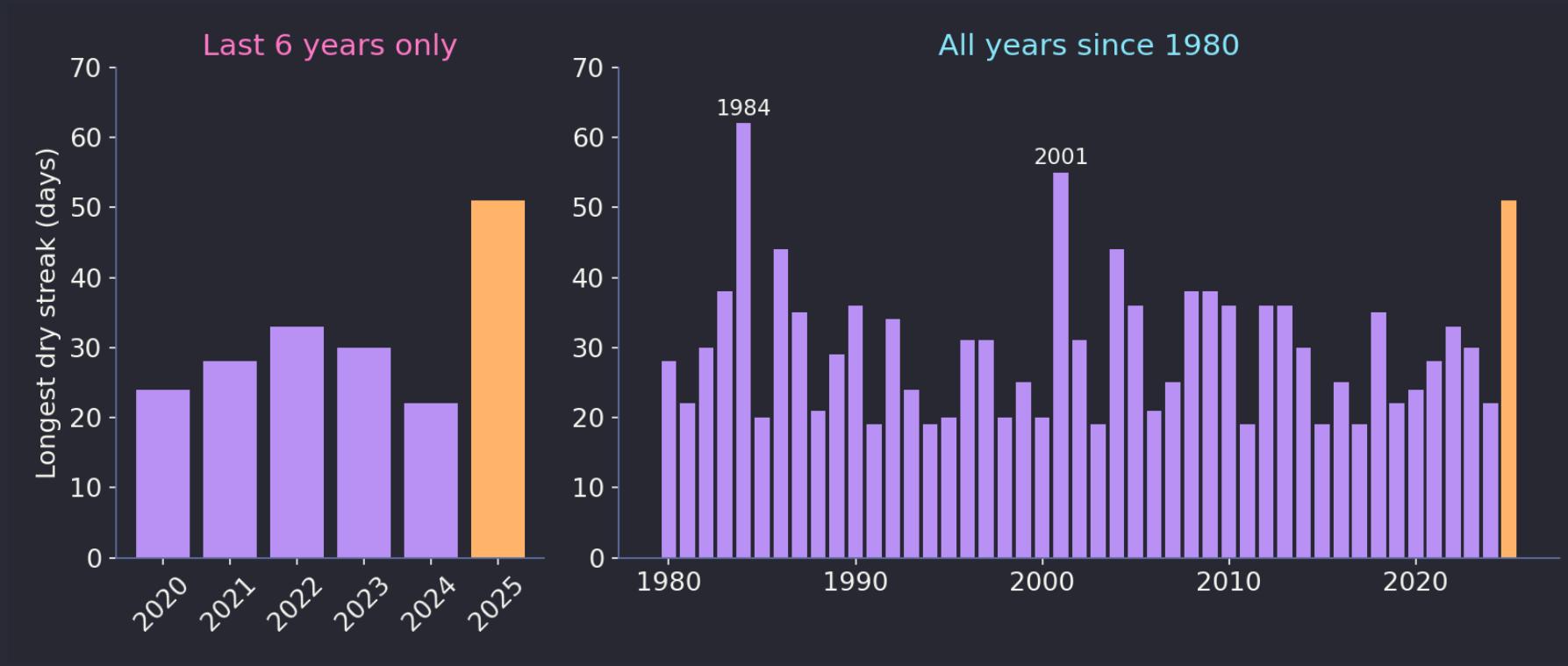
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2. Does the Measure Make Sense?

- Ask: **what exactly is being shown**, and is it the right measure?
  - Video example: a dollar cost shown as **how much space the cash would fill** — vivid, but not useful
- **3D and picture graphs** look good but **distort** size
  - Doubling an icon's height also doubles its width → looks **4×** bigger
- **Mixed data**: two different measures on one chart with **two y-axes**
  - Each axis can be stretched to make the lines "agree"

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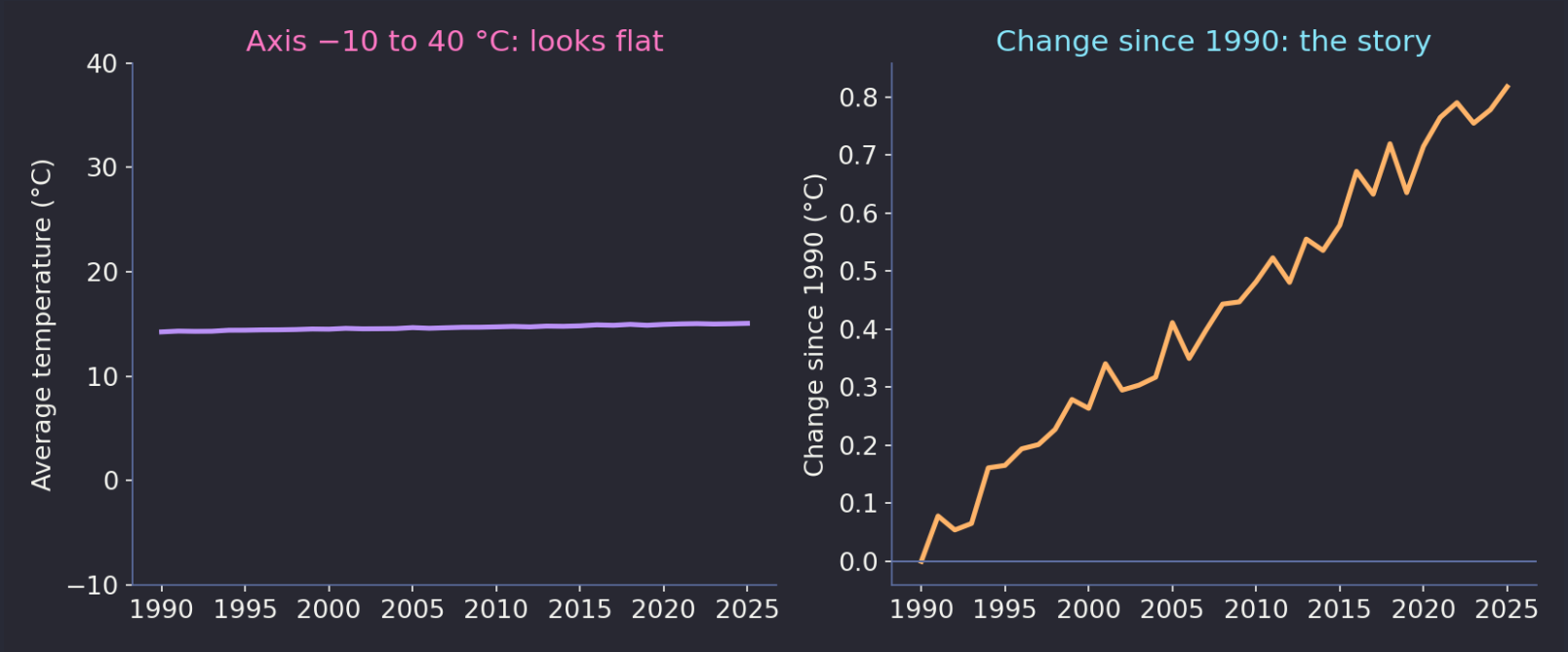
3. Cherry-Picking — What Was Left Out?



Illustrative data. Left: a short window makes 2025 look like a record. Right: it isn't.

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4. Context — Is the Scale Right for the Story?



Illustrative data. A scale that is too wide can hide a change that matters.

- Zooming **out** can mislead as much as zooming **in**
- Look for **ignored variables**: population, inflation, season

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5. Check the Source

- **Who made it** — and what do they gain if you believe it?
- Reputable outlets have **editors and fact-checkers**
- Ads, press releases and **sponsored studies** deserve extra suspicion
- Not sure? **Find the underlying data** before you share it

> *If you can't find the data behind the chart, you can't check the chart.*

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Careless or Dishonest — Same Result

Careless

- Default axis settings left as-is
- Short date range "because that's what I had"
- 3D style because it looked nice

Dishonest

- Axis chosen to exaggerate a gap
- Years dropped that spoil the story
- Units picked to sound dramatic

> *Readers can't tell which one it was — so **check your own charts** as hard as other people's*

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AI Gets Fooled Too

- A 2025 study showed **10 AI models** the same data drawn **honestly vs. misleadingly**
- **Most models changed their answer** — the data didn't change, only the drawing
- Tricks that fooled the most models:
  - **Upside-down y-axis** — all 10
  - **Stretched / squashed** chart shape — 9
  - **Y-axis not starting at zero** — 7
- Best model still missed some tricks

> *Run the checklist on the chart — not on the AI's summary of it*

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Checklist — Before You Believe (or Publish) a Chart

- ☐ **Y-axis**: where does it start? Bars from **zero**? Right way **up**?
- ☐ **X-axis**: even intervals, nothing skipped, labelled with units?
- ☐ **Measure**: is it the right thing to show? Any 3D or picture distortion?
- ☐ **Time range**: why does it start and end there? What's missing?
- ☐ **Scale & context**: too zoomed in, or too zoomed out?
- ☐ **Other variables**: per capita? adjusted for inflation?
- ☐ **Source**: who made it, and can I see the data?

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Key Takeaways

- Graphs are powerful because they are **fast** — that is also why they mislead
- Check the **axes** first: start, spacing, scale
- Ask what's **missing**: years, variables, context
- Ask **who** made it and **why**
- Apply the same checks to **your own** charts — and to **AI summaries** of charts

> *Slow down for ten seconds before you trust a picture of data*

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References

Topic	Source
Video	Gaslowitz, L. (2017). *How to spot a misleading graph* [Video]. TED-Ed.
Lesson page & quiz	TED-Ed. (2017). *How to spot a misleading graph — lesson*.
More examples & practice	Front Porch Math. *Misleading graphs* (TED-Ed "Dig Deeper" resource).
Lesson summary	Upworthy. *Don't be fooled: 5 ways to spot a misleading or careless graph*.
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Classic text	Huff, D. (1954). *How to Lie with Statistics*. W. W. Norton.
Data-ink & graphical integrity	Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). Graphics Press.
Charts in this deck	Illustrative data generated by SophiArch for teaching purposes.

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Thank You

- Questions?

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